



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2024-25(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: V

Programme: B. Tech Information Technology

Course: Digital Logic & Fundamentals of Microprocessor

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BEC33510	Course Outcome
CO1	To classify the working principles of various combinational circuits and their applications in different digital systems.
CO2	To analyze and recognize various sequential components utilized in the design of combinational circuits.
CO3	To develop the skills to design and validate the operation of various combinational and sequential circuits.
CO4	To provide knowledge on the 8085 microprocessor, covering its architecture, functionalities, addressing modes, port types, and practical applications.
CO5	To explore the use of Assembly language programming for the 8085 microprocessor, emphasizing its timing, control features in industrial embedded systems.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define K-map.	CO1	2	2
b.	What is Demultiplexer?	CO2	2	2
c.	Define counter.	CO3	2	2
d.	What is opcode?	CO4	2	2

e.	What is masking of interrupts?	CO5	3	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Convert binary to decimal and vice versa.	CO1	2	5
b.	Simplify Boolean expression using K-map.	CO1	3	5
c.	Prove De Morgan's theorem using truth table	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Design a full subtractor and explain.	CO2	2	5
b.	Explain working of comparator.	CO2	4	5
c.	Design 2-to-4 decoder using logic gates.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain D flip-flop with truth table.	CO3	2	5
b.	Design a MOD-10 counter.	CO3	3	5
c.	Convert JK flip-flop into D flip-flop.	CO3	4	5
Que. 5	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain instruction cycle of 8085.	CO4	3	5
b.	Describe different types of instructions with examples.	CO4	3	5
c.	Explain registers and flags in 8085.	CO4	4	5
Que. 6	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain vectored and non-vectored interrupts.	CO5	3	5
b.	Write assembly program to add two numbers.	CO5	4	5
c.	Explain timing diagram of I/O read cycle.	CO5	2	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create